## ✱ Feature selection :

- minimal setup features that are good enough for ✱ a
  ↳ use need minimal features to :
  (1) reduce computational complexity

• No. of samples << No. of features

S1 $x_1, x_2, x_3 \dots\dots x_{100,000} \rightarrow y^{(1)}$

S2 $x_1, x_2, x_3 \dots\dots x_{100,000} \rightarrow y^{(2)}$

S3 $x_1, x_2, x_3 \dots\dots x_{100,000} \rightarrow y^{(3)}$

✱ If more features, les
samples overfitting
✱ Ideal no. of samples
ML : 10 × No. of
DL : 10000 × No. of

## ✱ Strategies :

→ ✱ LASSO / L1 Regularisation
→ Forward Search
→ Backward Search
→ Recursive Feature Elimination (RFE)

precision and recall

Specificity

0.7

ROC

0

sensitivity

1.0

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1. Forward search :-

$x_1$ $x_2$ $x_3$ $x_4$ $x_5$

# Goal is to come down from $x$ no. of features to a smaller number

Step 1 :- $F = \{\phi\}$ # empty set F (no. features)

$M_1 = h_0(x) = \theta_0$ # the first model

Step 2 :- Add feature $x_1$ to F

$F = \{\phi, x_1\} \rightarrow M_1 \rightarrow AUC_1$ (scoring matrix)

Repeat this step, pick one feature at a time → this tells us how much that feature will increase the performance of model

$F = \{\phi, x_2\} \rightarrow M_2 \rightarrow AUC_2$

$F = \{\phi, x_3\} \rightarrow M_3 \rightarrow AUC_3$

$F = \{\phi, x_4\} \rightarrow M_4 \rightarrow AUC_4$

$F = \{\phi, x_5\} \rightarrow M_5 \rightarrow AUC_5$

Pick feature with maximum AUC

Let us say $x_2$ performed better - then commit $x_2$ to feature $F = \{x_2\}$ (final solution list)

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Step 3 : $M_1 = \{x_2, x_1\} \rightarrow AUC_1$
$M_2 = \{x_2, x_2\} \rightarrow AUC_2$
$M_3 = \{x_2, x_3\} \rightarrow AUC_3$
$M_4 = \{x_2, x_4\} \rightarrow AUC_4$
$M_5 = \{x_2, x_5\} \rightarrow AUC_5$

# Say $x_4$ gives better performance. Then combait it to the feature list
$F = \{x_2, x_4\}$

Do this until no more performance can be increased or when another feature is added, the performance is decreased. This is a good method but computationally very expensive.

(2.) Backward Search : $x_1, x_2, x_3, x_4, x_5$

- Take all the feature at once, then go on by dropping each feature one by one.

- If an important feature is dropped, the sorting metrics change accordingly.
  - Start with all the features $F = \{x_1, x_2, x_3, x_4, x_5\}$
  - remove / drop feature one by one.

(3.) RFE (similar to backward search) :
- "feature importance"
  - ↳ coefficient values (↑ value, ↑ importance)
  - ↳ probability scores

# instead of metrics we are looking at the prob. score.

→ Find which feature is most imp, drop it accordingly.
$F = \{x_1, x_2, \ldots x_5\}$

- Train a model, look at the scores for each feature.
- Remove features with the lowest scores

# At each step, we look at scores, and eliminate accordingly. ⇒ Recursive.